$- [S^0 (urea) + S^0 (H_2O)]$

$\Delta S_r^0 = [2 \times 192.5 + 213.5]$

$- [173.8 + 70]$

$\Delta S_r^0 = [598.5] - [243.8]$

$\Delta S_r^0 = 354.7 \text{ J mol}^{-1} \text{ K}^{-1}$

7.

Solution:

Given:

$T_b = 351 \text{ K}$

$\Delta H_{vap} = 39840 \text{ J mol}^{-1}$

$\Delta S_v = ?$

$$\Delta S_v = \frac{\Delta H_{vap}}{T_b}$$

$$\Delta S_v = \frac{39840}{351}$$

$$\Delta S_v = 113.5 \text{ J K}^{-1} \text{ mol}^{-1}$$

8.

Solution:

Given:

$\Delta H = -10 \text{ kJ mol}^{-1} = -10000 \text{ J mol}^{-1}$

$\Delta S = -20 \text{ J K}^{-1} \text{ mol}^{-1}$

$T = 300 \text{ K}$

$\Delta G = ?$

$\Delta G = \Delta H - T\Delta S$

$\Delta G = -10 \text{ kJ mol}^{-1} - 300 \text{ K} \times (-20 \times 10^{-3}) \text{ kJ K}^{-1} \text{ mol}^{-1}$

$\Delta G = (-10+6) \text{ kJ mol}^{-1}$

$\Delta G = -4 \text{ kJ mol}^{-1}$

At 600 K

$\Delta G = -10 \text{ kJ mol}^{-1} - 600 \text{ K} \times (-20 \times 10^{-3}) \text{ kJ K}^{-1} \text{ mol}^{-1}$

$\Delta G = (-10 + 12) \text{ kJ mol}^{-1}$

$\Delta G = +2 \text{ kJ mol}^{-1}$

The value of ΔG is negative at 300K and the reaction is spontaneous, but at 600K the value ΔG becomes positive and the reaction is non spontaneous.

EVALUATION

I Choose the best answer:

1. (b) ΔH
2. (d) decrease in free energy
3. (b) $q = 0$
4. (d) = 0
5. (a) $w = -\Delta U$
6. (d) $\frac{\text{mass}}{\text{volume}}$
7. (a) – 900 J
8. (b) negative
9. (b) – 67.6 kcal
10. (a) graphite is more stable than diamond
11. (d) – 462 kJ
12. (d) frictional energy
13. (d) $\Delta H < \Delta U$
14. (c) + 3 kJ
15. (a) – 2.48 kJ
16. (b) – 500 R
17. (d) $\frac{b-2a}{2}$

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